Internet programming

Internet programming is development of web based applications and websites that run on internet using technologies such as html, css, Js, mySQL & server side languages

It includes Front END development — designing user interface using html, css and Javascript

Back end development - processing user request storing data communicating with data base using programming languages like python, Java, PHP.

Data base:- It's stores and retrieves the data.

Ex:-

Online shopping website (Any online shopping amazon, flipkart, blinkit etc)

suppose a user wants to buy a mobile phone user opens the browser and opens the website user will search a mobile and adds it to the cart make the online payment.

-> Here Html creats the web page structure
-> css make page attractive
-> Javascript provides interactivity.
-> server side programming (python, Java, PHP) process the order,
* Data base

Features of internet programming

- It runs on the web browser
- works over internet
- Supports claint server communication
- Allows user interaction
- user data base to store the data and information
- can be accessed from any where with an internet connection

Examples of IP / applications

- e-commerce website
- online Banking
- social media platform
- email services
- online ticket Booking
- video streaming website

Internet – Internet is a global network of inter connected compater networks that enables communication and information around world wide.

WWW – world wide web.
It is also called as ocean of websites
www is a collection of inter connected web page and web sites that can be accessed over internet using web browser

Components of WWW

1) web browser – It is a software application which is use to access and display the web pages.
2) web server – It is a computer that stores web site and its files. which delivers the web pages to the users when requested
Ex:- apache, microsoft IIS.
3) web page :– It is the single document on www usually created with html.
ex:- Home page, contact page
4) web site :– collection of related web pages stored on web server under single domain name.
Ex:- amazon.com
5) URL – (Uniform Resource locater)
URL is the unique address of the web page.
ex:- https://www.amazon.com/about

Parts of URL

1) Protocol ~~https~~ (set of rules) = ~~ht~~ https
2) Domain name = www.amazon.com
3) specific web page = about

Http → Hypertext transfer protocol
→ It is the set of rules use to transfer web pages between web server & web browser.

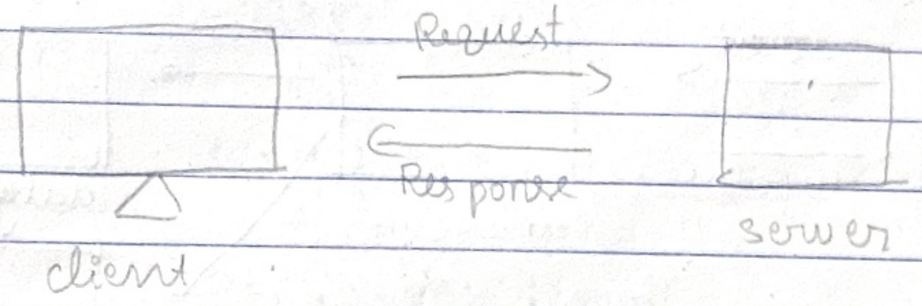
Difference between Http and https

http	https
- Hyper text transfer protocol	- Hyper text transfer protocol seared
- No data encryption	- It transfers data using encryption
- Data can be intersepted by hackers	- Data is protected by Hackers
- It can be used in the websites that do not handle sensitive information	- It can be used in the websites for sensitive information

Type of website
1) static
2) dynamic

Static website	dynamic website
→ content is fixed for every user	→ content changes dynamically
→ same content for every user	→ different users can see different content
→ Easy to develop	* more complex to develop
→ No database is required	→ data base is required.
→ Content changes only when if is manually updated	→ content usually changes based on user requirements

Client - server architecture (static)

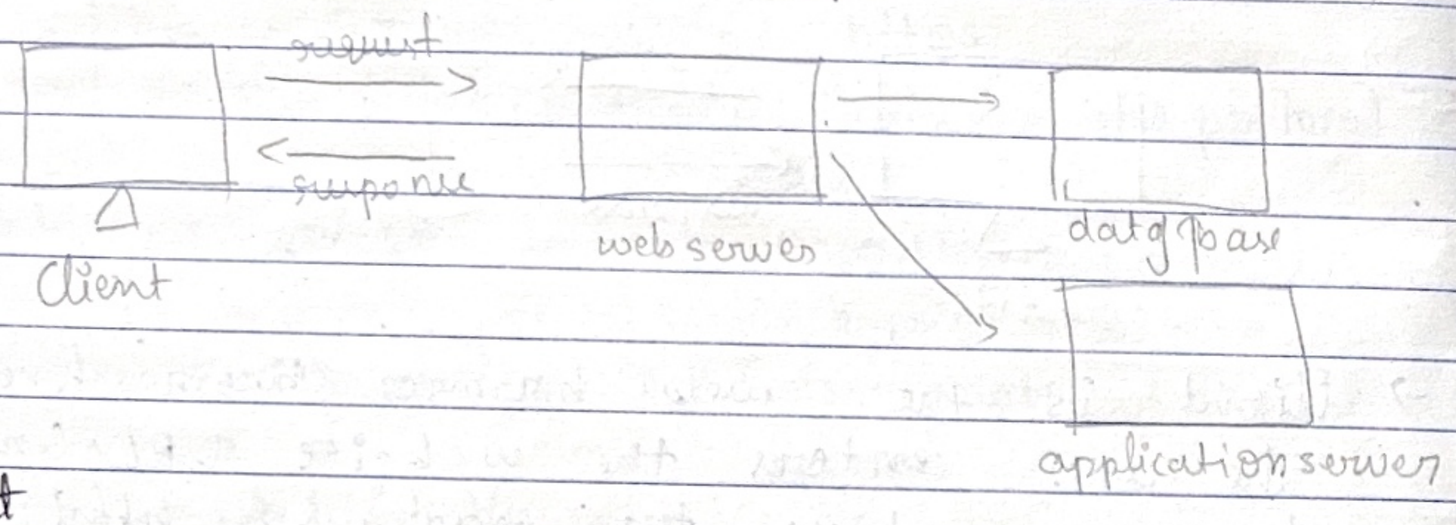


→ Client is the web browser (Chrome, firefox)
→ The user enters the website URL (request)
→ web server stores the readymade html, css, Java script files.
→ It simply sends the files to the client as a response
→ It doesnot need the data base
→ The browser displays the requested page to the user
Ex:- A college information website.

flow diagram

user request the page using browser
↓
The browser sents the request to the browser
↓
web server recieves the request
↓
web server sents the html/css/Javascript files to the browser
↓
Browser display it to the user.

Client server architecture (dynamic)



Client

The user browser sents a request

web server

web server recieves the request
→ web server forwards the request to the application server
→ Application server is the brain of the website
It contains the back end code.

Application server

→ It validates the user login executes the business logic and generates the web page dynamically.
→ It can read and update the data base.

Data base

→ Data base stores all the website data like user name password, ~~stau~~ student records, product, product details, ordered details, massages & reviews.
Ex:- MySQL, oracle

Flow diagram

user
↓↑
web server
↓↑
application server
↓↑
data base

Html

Hyper text markup language.

Hyper → Non linear navigation using hyper links

text → The content display on the web page

markup → Tags are to define the structure of the contents

language → It is the set of rules for creating a web page

Characters or features

→ simple and easy to learn
→ Platform independent
→ uses tags to structure the content
→ supports hyperlinks
→ supports multipedia
→ It works with css and Java script
→ supported by all web browser
→ It is not case sensitive

advantages of html
→ simple & easy to learn
→ platform idependent
→ It is free & open source
→ supported by all web browser

disadvantages
→ needs css for styling
→ html can't perform calculation / logic
→ need Java script for interactivity
→ It can't create dynamic websites by itself

html page structure
<!DOCTYPE html>
<html>
<head>
<title> something </title>
</head>
<body>
 Hello world
</body>
</html>

→ <!DOCTYPE html>
It specifies the browser that it is using html 5 version

<html> → Root element which contains entire html page

<head> → It contains the meta data / more information about the web page like title, character encoding, css links, Js links.

<title> - It specifies the title of the web page it is displayed on the browser tag

<body> - It contains all the visible contents.

⑤ html element
→ html element are the building blacks of the web page
→ They are represented by tags and define structure and content of the web page

element	discription	example
<html>	it is the root element	<html> ⋮ </html>
<head>	It contains the meta data.	<head> <title>---</title> </head>
<title>	It sets the page title.	<title> Hellow </title>
<body>	It is the visible contains of the web page.	<body> <h> Hellow </h> </body>

Types of element.
1) Paired element
2) Unpaired element

① Paired element
-> It should have opening and closing tags.
-> They contain content between tags
syntax:
<tag> content </tag>
Ex:-

② Unpaired element / selfclosing element / empty ~~tags~~ element
-> It donot have closing tags.
-> They donot have the content.
syntax:-
<tag>
ex: <meta charset = "UTF-8"> -> supports many language

 -> to break the line
<hr> -> It breaks the line & gives the
<img> -> used to insert the image
<link> -> used to links the css file
<source> ->

Heading.
-> html heading used to define title and sub titles and sub headings of a web page.
-> html provides 6 headings from <h1> to <h6>
-> <h1> be the most important heading (largest)
-> <h6> is the least important heading (smallest)

Example program
```
<!doctype html>
<html>
<head>
<title> heading </title>
</head>
<body>
<h1> welcome </h1>
<h2> welcome </h2>
<h3> welcome </h3>
<h4> welcome </h4>
<h5> welcome </h5>
<h6> welcome </h6>
</body>
</html>
```

③ html comments
-> html comments add notes or explation of the source code.
syntax:-
<!-- comment -->
-> It is not display on the web browser.

* html text formating
-> It is used to change the appearnce of text
-> such as making it bold, italic, highlighted superscript or subscript.
Ex:- H ₂ O = H_2O

tag	discription
<b>	make the text bold.
	Indecate important text (usually bolds)
<i>	make text italic.
	make text italic only.
<u>	underline the text.
<mark>	highlights the text
<del>	~~text~~
<ins>	shows the underline text.
<sub>	shows the subscripted text
<sup>	shows the superscripted text

Ex)

```
<!doctype html>
<html>
<head>
<title> formating </title>
</head>
<body>
<i> Hi </i>
<strong> Hi </strong>
<b> H <sub> 2 </sub> O <b>
<em> Hi </em>
<u> Hi </u>
<mark> Hi </mark>
<del> Hi </del>
<ins> a <sup> 2 </sup> </ins>
</body>
</html>
```

· Create a html file page using tags

a) paragraph containg text · all ~~details~~ is that ~~not~~ letters are not gold in bold face and italisize the text.

```
<!doctype html>
<html>
<head>
<title> Paragraph </title>
</head>
<body>                    all that letters are not gold
<P> <b> <i> Hi, welcome to IP </i> </b> </P>
</body>
</html>
```

b) create a equation $x = \frac{1}{3}(y_1^2 + z_2^1)$

```
<!doctype html>
<html>
<head>
<title> equation </title>
</head>
<body>
<P> x = Y3 ( y <sub> 1 </sub> <sup> 2 </sup> + z <sub> 2
</sub> <sup> 1 </sup> ) </P>
</body>
</html>
```

html paragraph.

In html paragraph is created using <P> tag

→ It is the paired tag.

→ Browser automatically adds space before & after each paragraph.

Ex:-

```
<!doctype html>
<html>
<head>
<title> paragraph </title>
</head>
<body>
<P> This a paragraph </P>
<P> This is another paragraph </P>
</body>
</html>
```

html list.

It is used to display related iteams in organised way

Types

1) ordered list →
2) Un order list →
3) discripsion list → <dl> </dl>

Ordered list

Ex: <!doctype html>
<html>
<head>
<title> movies </title>
</head>
<body>
 my fav movies
<ol type="I">
 Jagamalla
 24
 Jagnam
 milana

</body>
</html>

types=
A
a
i
I
reversed

2) Unorder list

```
Ex. <body>
<b><em> My fav movies </em></b>
<ul type="square">
<li>  </li>
<li>  </li>
<li>  </li>
<li>  </li>
</body>
```

1) Create unorded list of 5 fruits and order list of three flowers

→ <!doctype html>
<body>
 5 fruits

 apple
 banana.
 cherry
 data
 papaya

</body>

<body>
 3 flowers

 Rose
 Jasmin
 Lilly

</body>

<body>
<a! type="I">
 list of flowers

 ——
 ——
 ——

</body>

3) discripsion list (dl)
It is used to display that terms and discripsion.
→ tags used <dl> <dt> <dd>
discripsion term, discripsion defination

Ex: <body>
<dl>
<dt> HTML </dt>
<dd> Hyper text markup language </dd>
<dt> CSS </dt>
<dd> cascode style sheet </dd>
</dl>
</body>

* Attribute
Html attribute provides additional information about the tag or the element.
→ It is writern inside the opening tag.
Ex:- <ol type="I">

→ Table in html.
Tags.
<th> </th> -> table heading
<tr> </tr> -> table row
<td> </td> -> Table data.
<table> </table> -> Table
<caption> </caption> -> .caption it adds title to the table.

Ex: <!Doctype html>
<html>
<head>
<title> table </title>
</head>
<body>
<table>
<caption>

Ex. <!Doctype html>
<html>
<body>
<table border="1">
<tr>
<th> Name </th>
<th> age </th>
</tr>
<tr>
<td> xyz </td>
<td> 98 </td>
</tr>
<tr>
<td> ppr </td>
<td> 97 </td>
</tr>
</table>
</body>
</html>

Ex:- <Body>
<table border="1">
<caption> clg details </caption>
<tr>
<th> ~~name~~ clg </th>
<th> ~~city~~ </th>
</tr>
<tr>
<td> PES </td>
<td> shmg </td>
</tr>
</table>
</body>

Cell padding -> It provides space inside a cell.

Cell spacing -> space between the cell.

Bgcolor =>

Ex:-
Write a html program to create a table showing rollno, name, marks and the total.

<!Doctype html>
<html>
<body>
<table border="2" cellpadding="10" cellspacing="5"
width="80%" Height="300" allign="center"
Bgcolor="blue" >
<caption><B> student marks report </B></caption>
<tr>
<th> ~~Rowspan~~ <th rowspan="2"> name </th>
<th rowspan="2"> roll no </th>
<th ~~rowspan~~ ~~colspan~~ colspan="3"> marks </th>
<th rowspan="2"> total </th>
</tr>
<tr>
<td> html </td>
<td> CSS </td>
<td> JS </td>
</tr>
<tr>
<td> Ammu </td>
</tr>
</table>
</body>
</html>

```
<td> 1 </td>
<td> 98 </td>
<td> 95 </td>
<td> 97 </td>
<td> 290 </td>
</tr>
<tr>
<td> chand </td>
<td> 8 </td>
<td> 95 </td>
<td> 85 </td>
<td> 90 </td>
<td> 540 </td>
</tr>
<tr>
<td> Priya </td>
<td> 3 </td>
<td> 90 </td>
<td> 90 </td>
<td> 90 </td>
<td> 270 </td>
</tr>
<tr colspan="">
<tr>
<td colspan="5"> Grand total </td>
<td> 830 </td>
</tr>
</table>
</body>
</html>
```

Image tag (<img>)

It is used to display the images on a web page.

Syntax:- <img src="flower.jpg" alt="flower">

→ unpaired element

Attribute	Description
Src(Source)	Specifies the image file path or URL
~~alt(alternative)~~	
alt(alternative)	It is the alternative text diplay if the image cannot load.
Height	resize the image height
width	resize the image width.
title	It shows a tooltip when a mouse is over the image

Ex:-
```
<!doctype html>
<html>
<body>
<img scr="winners.jpeg" alt="winners" Height="10"
     width="10" title="winners BCA">
<figcaption> "college" </figcaption>
</body>
</html>
```

Back ground image

Is an image displayed beihand the containt of the web page

→ It is add using CSS

Syntax:-

```
background-image: url("filename.jpeg");
```

properties	discripsion
Background_image	to insert the background image
Background_repeat	repeates or stop repeating the image
Background_size	controls the size of the image
Background_position	to set the position of the image
Background attachment	

Ex- <!Doctype html>
<html>
<head>
<style>
body {
background_image: URL ("flower.JPG");
background_repeat: No-repeat; → repeat
background_size: cover; contain → 100% 100%; 100 50%
background_position: center; top, left, bottom, right
background_attachment: scroll;
background_color: light blue;
}
</style>
</head>
<body>
—
—
</body>
</html>

Ex program
<head>
<style>
img {
width: 180PX;
height: 180PX;
margin: 10PX;
}
.round {
border: 5px dotted red;
border-radius: 50 PX;
}
.circle {
border: 5PX double blue;
border-radius_radius: 50%
}
</style>
</head>

Division tag in html
It is a block level container use to group html elements together
It helps organize a web page and apply same css style (inside) to multiple elements.
~~system~~ syntax:-

content

Html colors

Html provides color to the text, to the background and to the webpage.

syntax

```
<font color = "colorname">
</font>
```

Hexadecimal values for color

Red - #ff0000

green - #008000

Blue - #0000ff

Yellow - #ffff00

Black - #000000

white - #ffffff

background

```
<body bgcolor = 'colorname' >
```